

MH1101 Tutorial 11 (Week 12)

Reference: Topic 6.1, 6.2 (Lecture Notes)

1. Find the radius of convergence and interval of convergence of the series.

(a)
$$\sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)2^n} (x-1)^n$$

(b)
$$\sum_{n=1}^{\infty} \frac{\sqrt{n}}{8^n} (x+6)^n$$

(c)
$$\sum_{n=2}^{\infty} \frac{b^n}{\ln n} (x-a)^n, b > 0$$

(d)
$$\sum_{n=1}^{\infty} \frac{x^n}{1 \cdot 3 \cdot 5 \cdots (2n-1)}$$

2. Suppose that the power series $\sum c_n(x-a)^n$ satisfies $c_n \neq 0$ for all n .

Show that if $\lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right|$ exists, then it is equal to the radius of convergence of the power series.

3. Suppose the series $\sum c_n x^n$ has radius convergence R . What is the radius of convergence of the power series $\sum c_n x^{2n}$? Justify your answer.

4. Find a power series (in x) representation for the function by manipulating a geometric series and determine the interval of convergence.

(a)
$$f(x) = \frac{4}{2x+3}$$

$$(b) f(x) = \frac{x^2}{x^4 + 16}$$

$$(c) f(x) = \frac{x-1}{x+2}$$

5. By differentiating or integrating certain power series, find a power series (in x) representation for the function and determine the radius of convergence.

$$(a) f(x) = \ln(5-x)$$

$$(b) f(x) = \left(\frac{x}{2-x} \right)^3$$

6. (Some applications of power series)

(a) Use the first three terms of a power series to evaluate $\int_0^{0.4} \ln(1+x^4) dx$.

(b) Find the sum of the series

$$\sum_{n=2}^{\infty} \frac{n(n-1)}{2^n}.$$

Hint: Evaluate $\sum_{n=2}^{\infty} n(n-1)x^n$, $|x| < 1$, using a power series. Recall that in Example 6.9 of Chapter 6, we have $\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} nx^{n-1}$, $|x| < 1$.

Answer Keys. **1.** (a) $(-1, 3]$ (b) $(-14, 2)$ (c) $\left[a - \frac{1}{b}, a + \frac{1}{b}\right)$ (d) $(-\infty, \infty)$

3. $\sqrt{R}$ **4.** (a) $\sum_{n=0}^{\infty} (-1)^n \frac{2^{n+2}}{3^{n+1}} x^n$ (b) $\sum_{n=0}^{\infty} (-1)^n \frac{x^{4n+2}}{2^{4n+4}}$ (c) $-\frac{1}{2} - \sum_{n=1}^{\infty} (-1)^n \frac{3x^n}{2^{n+1}}$

5. (a) $C - \sum_{n=1}^{\infty} \frac{x^n}{5^n n}$ (b) $\sum_{n=0}^{\infty} (n+2)(n+1) \frac{x^{n+3}}{2^{n+4}}$ **6.** (a) 0.002034 (b) 4